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SECTION 01

Executive Summary

This document specifies the technical architecture for the **Real-Time Call Analysis (RTCA) Module** — a cross-domain capability extension of the NOUS Platform that enables live cognitive-linguistic surveillance during telephone calls, telehealth sessions, video consultations, and other real-time audio interactions. The RTCA Module transforms every conversation into a clinical data stream, applying the full NOUS processing pipeline — REMIEL extraction, NOST oscillatory mapping, and NOSS 12-sector classification — in a continuous, streaming mode with sub-second latency.

The foundational insight is straightforward: NOUS already processes language to detect cognitive, neurological, cardiovascular, oncological, and prenatal health signatures. Currently, this processing operates on collected samples — recorded narratives, written responses, submitted voice diaries. The RTCA Module eliminates the collection step entirely. Every phone call with a patient, every telehealth check-in, every crisis hotline interaction, every post-game interview becomes a live assessment opportunity — analyzed in real time, with alerts generated while the conversation is still in progress.

The clinical implications are profound. A primary care physician on a routine telehealth call receives an on-screen alert midway through the conversation: "*Elevated S4 trajectory with prosodic-autonomic features — cardiac evaluation recommended.*" The physician didn't ask about the heart. The patient didn't mention chest pain. But the language betrayed what the body could not conceal. The doctor adjusts the conversation in real time. A referral is placed before the call ends.

The RTCA Module does not introduce new science. It introduces a new modality of deployment for the existing NOUS architecture — one that converts passive infrastructure (phone systems, telehealth platforms, call center software) into active clinical surveillance instruments.

> **Design Principle:** The RTCA Module is not a separate product. It is a deployment mode of the existing NOUS processing pipeline, adapted for streaming audio input with real-time output rendering. Every capability that NOUS provides in batch mode — 12-sector NOSS classification, trajectory analysis, differential engine, clinical alert generation — operates identically in the streaming context. The science is unchanged. The delivery mechanism is new.

SECTION 02

Strategic Rationale

2.1 Market Opportunity

Real-time conversational AI in healthcare represents one of the fastest-growing segments in digital

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health. The RTCA Module addresses multiple converging market forces:

Market Driver	Detail
Telehealth Permanence	Post-pandemic telehealth utilization has stabilized at 15–20% of all outpatient visits, representing ~300 million clinical conversations annually in the U.S. alone. Each conversation is a potential NOUS assessment opportunity currently going unanalyzed.
Call Center Scale	Health plan call centers, nurse triage lines, and patient navigation services handle an estimated 1.2 billion calls per year in the U.S. healthcare system. These interactions contain clinically relevant language that is captured, stored, and never analyzed.
Crisis Line Demand	The 988 Suicide & Crisis Lifeline processed 5.5 million contacts in 2024 — a 40% increase year-over-year. Crisis counselors lack real-time objective risk stratification tools; NOUS provides exactly this capability.
Ambient Clinical Intelligence	The ambient AI documentation market (Nuance DAX, Abridge, Nabla) has validated clinician appetite for conversational AI during encounters. RTCA extends this paradigm from documentation to diagnostic intelligence.
Sports & Defense Real-Time	Post-game media sessions, mission debriefs, and operational check-ins represent high-value, time-constrained interactions where batch processing introduces unacceptable delays. Real-time output is operationally necessary.

2.2 Competitive Differentiation

Existing real-time conversational AI in healthcare focuses narrowly on documentation — transcribing encounters and generating clinical notes. No platform performs live multi-domain cognitive-linguistic surveillance during calls. The competitive landscape reveals a decisive gap:

Capability	Ambient AI (Nuance/Abridge)	Call Analytics (CallMiner/Cogito)	NOUS RTCA
Transcription	Yes	Yes	Yes (via REMIEL)
Sentiment Analysis	Basic	Yes	Superseded by NOSS
Clinical Note Generation	Yes	No	Not primary function
Cognitive Assessment	No	No	12-sector NOSS classification
Prosodic-Autonomic Analysis	No	Minimal	Full Pathway 3 cardiac detection

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Multi-Domain Surveillance	No	No	5 clinical domains simultaneous
Trajectory Alerting	No	No	Real-time trajectory comparison
Scientific Foundation	NLP/LLM-based	NLP-based	NOST — proprietary oscillatory theory
Patent Protection	Standard NLP	Standard NLP	35 pending applications

The competitive moat is structural: sentiment analysis identifies how someone feels about a conversation; NOUS identifies what is happening in their brain. No ambient AI platform detects early cardiac autonomic dysregulation from speech rhythm changes. No call analytics engine classifies neurodegenerative trajectories from lexical shift patterns. The RTCA Module brings NOUS's unique scientific capabilities into the real-time domain — a market position that cannot be replicated by adding features to existing transcription or sentiment platforms.

2.3 Revenue Impact

The RTCA Module transforms NOUS from a per-assessment tool into a per-minute or per-encounter platform, fundamentally expanding the revenue model:

> **Revenue Model Expansion:** Current NOUS pricing is per-assessment or per-practitioner subscription. RTCA enables per-minute processing fees for high-volume call center deployments ($0.08–$0.25/minute), per-encounter licensing for telehealth integration ($3–$8/encounter), and enterprise platform fees for health system-wide deployment ($250K–$1.5M/year). Conservative Year 3 RTCA revenue estimate: $8.5M–$14.0M, representing incremental revenue above existing NOUS projections.

SECTION 03

System Architecture

3.1 Architectural Overview

The RTCA Module operates as a streaming adapter layer that connects real-time audio sources to the existing NOUS processing pipeline. It does not replace or duplicate core NOUS engines — it provides a continuous-input interface to them, along with real-time output rendering and alert delivery mechanisms.

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RTCA MODULE — STREAMING ARCHITECTURE

AUDIO INGESTION LAYER

SIP/VoIP Tap | WebRTC Stream | Telehealth API | PSTN Bridge | Mobile SDK | Conference Bridge

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PRE-PROCESSING LAYER

Noise Suppression | Speaker Diarization | Voice Activity Detection | Channel Separation | Acoustic Normalization

▼

STREAMING REMIEL ENGINE

Streaming ASR | Live Lexical Extraction | Rolling Syntactic Analysis | Continuous Prosodic Monitor | Real-Time Temporal Features | Streaming Semantic Coherence

▼

LIVE NOST/NOSS ENGINE

Windowed Oscillatory Mapping (δ θ α β γ) | Rolling NOSS 12-Sector Scoring | Real-Time Baseline Comparison | Live Differential Engine

▼

REAL-TIME ALERT & OUTPUT LAYER

Live Clinician Dashboard | In-Call Alert Overlay | WebSocket Push Notifications | Post-Call Summary Generator | EHR Auto-Documentation

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3.2 Key Architectural Decisions

3.2.1 Windowed Processing vs. Frame-by-Frame

Linguistic analysis requires sufficient context to produce meaningful features. Individual words or sub-second audio frames cannot generate reliable NOSS sector activations. The RTCA Module employs a **sliding window architecture** with configurable overlap:

WINDOW CONFIGURATION	
Primary Analysis Window	**30 seconds (rolling)**
Window Overlap	**15 seconds (50%)**
Prosodic Micro-Window	**5 seconds (for F0/rhythm)**
NOSS Update Frequency	**Every 15 seconds**
Minimum Utterance for Scoring	**~50 words or 20 seconds of speech**
Confidence Threshold	**Increases with conversation duration**

This design ensures that NOSS scores are updated approximately every 15 seconds during active speech, providing near-real-time cognitive surveillance while maintaining the statistical reliability that clinical application demands. Early windows (first 60 seconds) produce provisional scores with lower confidence; as the conversation progresses and the analysis window accumulates context, confidence rises to match batch-mode accuracy.

3.2.2 Speaker-Aware Processing

In clinical conversations, at least two speakers are present — the patient and the clinician. The RTCA Module must isolate the patient's speech for cognitive assessment while optionally analyzing the clinician's communication patterns for quality metrics.

- **Speaker Diarization:** Real-time speaker identification and separation using voiceprint enrollment (if available) or unsupervised clustering for unknown speakers. The system requires approximately 10 seconds of speech to establish stable speaker models.
- **Target Speaker Designation:** For clinical use, the system allows pre-designation of the "analysis target" (typically the patient). In multi-party calls (e.g., family conferences), multiple targets can be tracked independently.
- **Channel Separation:** Where stereo or multi-channel audio is available (most VoIP and conference systems), hardware-level channel separation provides superior speaker isolation to algorithmic diarization.

3.2.3 Edge-Cloud Hybrid Processing

To minimize latency and PHI exposure, the RTCA Module employs a hybrid architecture:

- **Edge Processing (on-device or on-premise):** Audio pre-processing (noise suppression, VAD, channel separation), initial prosodic feature extraction (F0, rhythm metrics, voice quality), and preliminary speaker diarization. Extracted features — not raw audio — are transmitted to the cloud processing layer.
- **Cloud Processing:** Full REMIEL extraction pipeline, NOST oscillatory mapping, NOSS scoring,

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trajectory comparison, differential engine, and alert generation. This layer accesses patient longitudinal data (prior assessments, established baselines) necessary for trajectory analysis.

- **Feature-Only Transmission:** In the highest-security configuration, raw audio never leaves the edge device. Only extracted numerical feature vectors (approximately 2KB per analysis window) are transmitted, dramatically reducing bandwidth requirements and PHI exposure.

SECTION 04

Streaming REMIEL Pipeline

The core technical innovation of the RTCA Module is the adaptation of the REMIEL Engine from batch processing to streaming mode. Each of REMIEL's five extraction domains must operate on partial, continuously updating input:

4.1 Streaming Lexical Extraction

Real-time automatic speech recognition (ASR) produces a streaming transcript with word-level timestamps. The lexical extraction module operates on a rolling buffer of recognized words, computing:

- **Rolling Type-Token Ratio:** Vocabulary diversity computed over the current 30-second window, updated with each new word. Sudden drops in TTR within a single conversation segment may indicate processing speed degradation or acute cognitive load.
- **Live Emotional Valence:** Positive/negative word ratio computed continuously, enabling detection of valence shifts that correspond to topic-triggered cognitive-emotional activation.
- **Absolutist Language Monitoring:** Real-time flagging of absolutist markers ("always," "never," "completely") — elevated rates correlate with suicidal ideation and are detectable within 30 seconds of onset.
- **Word-Finding Difficulty Detection:** Algorithmic identification of circumlocution, semantic paraphasia, and tip-of-the-tongue patterns in the live transcript — early markers of neurodegenerative and cerebrovascular pathology.

4.2 Streaming Syntactic Analysis

Syntactic features require complete clause structures for reliable analysis. The streaming parser operates on sentence boundaries detected by the ASR engine:

- **Incremental Sentence Complexity:** Mean clause depth, subordination ratio, and verb complexity computed per-sentence and averaged over the rolling window.
- **Syntactic Simplification Detection:** Real-time comparison of current syntactic complexity

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against the patient's established baseline (if available) or population norms. Progressive simplification during a conversation may indicate fatigue, cognitive decompensation, or acute neurological event.

- **Fragmentation Monitoring:** Detection of incomplete utterances, sentence abandonments, and grammatical breakdowns that may indicate acute confusion, TIA, or dissociative episode.

4.3 Continuous Prosodic Monitor

The prosodic channel operates on the shortest window (5 seconds) because acoustic features are extractable from brief speech segments. This is the highest-frequency data stream in the RTCA Module and provides the earliest available signal for cardiac-autonomic surveillance (Domain 3, Pathway 3):

- **Live F0 Tracking:** Continuous fundamental frequency extraction with 10ms frame resolution. Reduced F0 variability is the earliest prosodic marker of cardiac autonomic dysregulation — detectable in real time.
- **Speech Rhythm Index:** Pairwise Variability Index (PVI) computed over 5-second micro-windows. Rhythmic flattening indicates autonomic changes; rhythmic irregularity may indicate neurological disruption.
- **Respiratory Pattern Estimation:** Breath group duration and inter-breath interval estimation from the audio stream, providing non-contact respiratory rate monitoring and respiratory-cardiac coupling assessment.
- **Voice Quality Metrics:** Jitter, shimmer, and harmonics-to-noise ratio (HNR) computed continuously. Acute changes in voice quality during a single call may indicate emotional arousal, laryngeal tension changes, or autonomic shifts.

> **Clinical Significance — Prosodic Primacy:** The prosodic channel is uniquely suited to real-time analysis because it produces reliable features from the shortest speech segments and because the cardiovascular signatures it detects (Pathway 3: autonomic dysregulation) are the most time-critical. A patient experiencing cardiac autonomic deterioration during a telehealth call produces prosodic changes that RTCA can detect in the first 20 seconds of speech — potentially the fastest non-invasive cardiac alert mechanism available.

4.4 Real-Time Temporal Features

Processing speed indicators are inherently temporal and naturally suited to streaming analysis:

- **Live Response Latency:** In conversational contexts, the delay between a question's completion and the patient's response provides a real-time processing speed proxy. Extended latencies may indicate cognitive load, word-finding difficulty, or acute processing speed impairment.

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- **Pause Distribution Analysis:** Classification of pauses as filled (hesitations), unfilled (cognitive), or respiratory. Abnormal pause patterns emerge within single conversations and are among the most sensitive real-time markers of acute cognitive change.
- **Speech Rate Variability:** Words-per-minute and syllables-per-second computed over rolling windows. Progressive speech rate deceleration within a single call may indicate acute fatigue, medication effect, or neurological event.

4.5 Streaming Semantic Coherence

Semantic features require the most context and therefore operate on the longest effective window. The streaming semantic engine computes:

- **Rolling Coherence Score:** Sentence-to-sentence semantic similarity computed using embedding-based coherence metrics. Requires a minimum of 4–5 sentences (~30 seconds of speech). Coherence degradation during a conversation is a high-specificity marker for acute confusion, dissociative episodes, or psychotic decompensation.
- **Topic Drift Monitoring:** Algorithmic tracking of semantic distance from the conversation's nominal topic. Excessive tangentiality detected in real time may indicate disordered thought process.
- **Referential Clarity:** Pronoun resolution accuracy and referent tracking in the streaming transcript. Degraded referential clarity correlates with working memory impairment and is detectable from the first 60 seconds of conversation.

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SECTION 05

Clinical Applications by Domain

The RTCA Module applies across all five NOUS clinical domains and the three intelligence modules. The clinical value is domain-specific:

5.1 Domain 1 — Mental Health

The most immediately deployable RTCA application. Mental health assessment is already conversational; RTCA adds an objective computational layer to every clinical conversation.

Application	Mechanism	Clinical Output
Crisis Line Triage	Real-time NOSS sector activation during 988/crisis calls. S3 (psychotic), S1 (mood), S2 (anxiety), S10 (substance) scoring in the first 60 seconds.	Objective risk score displayed to crisis counselor during call. Automated escalation if scores exceed threshold.

Therapy Session Monitoring	Longitudinal comparison of NOSS profiles across therapy sessions conducted via telehealth. Treatment response measured objectively.	Session-by-session trajectory visualization. Therapist receives post-session cognitive change summary.
Psychiatric Emergency Detection	Acute changes in coherence (semantic), thought organization (syntactic), and reality testing (content) markers during any call.	RED alert to attending clinician if acute psychotic decompensation or suicidal crisis indicators detected.
Substance Use Detection	S10 activation from speech pattern analysis — slurred speech, prosodic irregularity, lexical changes consistent with acute intoxication or withdrawal.	Real-time substance influence flag during clinical encounters.

5.2 Domain 2 — Cerebrovascular

The RTCA Module enables the most clinically urgent application in the cerebrovascular domain: real-time detection of transient ischemic attack (TIA) or stroke during a phone call.

> **Acute Event Detection:** If a patient on a telehealth call experiences a TIA or stroke in progress, the RTCA Module detects the acute linguistic disruption — sudden word-finding failure, syntactic fragmentation, prosodic asymmetry — within seconds. This triggers a RED alert with the message: "*Acute cerebrovascular event signature detected — recommend immediate emergency evaluation.*" The treating physician receives this alert while still on the call, enabling immediate activation of stroke protocol. In this scenario, RTCA may be the fastest-available stroke detection system outside a monitored hospital setting.

5.3 Domain 3 — Cardiovascular

Cardiac surveillance benefits most from the RTCA Module's continuous prosodic monitoring capability:

- **Continuous Prosodic-Autonomic Monitoring:** Every telephone call with a cardiac patient becomes an autonomic function assessment. Reduced pitch variability, altered speech rhythm, and changed respiratory patterning are detected in real time and compared against the patient's established prosodic baseline.
- **Atrial Fibrillation Detection:** Patients with known or suspected AF produce prosodic signatures reflecting cardiac rhythm irregularity. RTCA can flag AF-consistent prosodic patterns during routine calls, prompting rhythm monitoring referral.
- **Heart Failure Decompensation:** Progressive prosodic deterioration across serial calls (trajectory analysis) or acute prosodic changes within a single call may indicate cardiac decompensation — providing a non-invasive, passive early warning system.

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5.4 Domain 4 — Oncology

For oncology patients, RTCA enables treatment-cycle-aware cognitive monitoring during routine clinical communication:

- **Post-Cycle Assessment Calls:** Oncology nurses typically call patients 48–72 hours after chemotherapy to assess side effects. RTCA automatically assesses cognitive-linguistic function during these already-scheduled calls, producing a neurotoxicity score without any additional patient burden or clinical workflow change.
- **Immunotherapy Safety Monitoring:** During scheduled telehealth check-ins, RTCA runs continuous surveillance for the subacute progressive cognitive decline that characterizes neurological immune-related adverse events (irAEs). Detection during a routine call can trigger urgent neurology referral hours or days earlier than symptom-based detection.

5.5 Domain 5 — GENESIS (Prenatal)

Prenatal RTCA applications leverage the Perinatal Mental Health Engine during routine obstetric telehealth contacts:

- **Perinatal Depression Screening:** Every telehealth visit or nurse check-in call during pregnancy becomes a continuous perinatal depression and anxiety assessment. Subtle cognitive-linguistic markers of prenatal depression are detected without requiring the patient to complete a questionnaire or self-report.
- **Preeclampsia Early Warning:** Prosodic-autonomic features associated with hypertensive cardiovascular changes may manifest during routine obstetric phone calls, enabling earlier clinical suspicion of developing preeclampsia.

5.6 Intelligence Modules — THANATOS, ORACLE, AEGIS

Module	RTCA Application
THANATOS	Mortality risk scoring during inpatient rounds, palliative care consultations, and ICU family conferences — providing real-time prognostic signals to inform care decisions.
ORACLE	Pharmacological response assessment during post-medication follow-up calls — detecting cognitive side effects or treatment response in real time.
AEGIS	Live cognitive fitness assessment during operational check-ins — military readiness evaluations, pilot fitness assessments, and executive cognitive screening conducted via standard communication channels.

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Deployment Scenarios

6.1 Telehealth Platform Integration

The primary deployment vector. RTCA integrates with major telehealth platforms (Doxy.me, Zoom for Healthcare, Teladoc, Amwell) via WebRTC audio stream access with clinician and patient consent:

TELEHEALTH INTEGRATION SPECIFICATION

Integration Method	**WebRTC media stream API**
Audio Format	**PCM 16-bit, 16kHz mono per channel**
Latency Budget	**<500ms ASR, <2s NOSS update**
Clinician Interface	**Side-panel overlay or separate dashboard**
Patient Consent	**Pre-session digital consent with opt-out**
Post-Session Output	**PDF summary + FHIR observation resource**

6.2 Call Center Deployment

High-volume deployment for health plan triage lines, nurse advice lines, and patient navigation services. RTCA processes calls in real time and provides cognitive-linguistic assessment to the agent:

- **Population-Level Screening:** Every inbound call is passively screened for cognitive, neurological, and psychological markers — converting the call center from a reactive service channel into a proactive surveillance instrument.
- **Triage Enhancement:** RTCA provides real-time risk scores that augment the nurse's clinical judgment during triage calls. A caller presenting with "headache" who simultaneously displays acute S4 trajectory changes is escalated immediately.
- **Quality Metrics:** Aggregated NOSS data across call populations enables health plans to identify undetected cognitive health trends in their member populations.

6.3 Crisis Line Integration

The highest-acuity deployment scenario. RTCA provides crisis counselors with real-time objective risk stratification:

> **988 Deployment Model:** During a crisis call, RTCA displays a live risk dashboard to the counselor showing S1 (mood), S2 (anxiety), S3 (psychotic), and S10 (substance) activation levels alongside prosodic distress markers. The system generates a composite crisis severity score and, if scores

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exceed a configurable threshold, automatically alerts a clinical supervisor. Post-call, a comprehensive NOSS assessment is generated and attached to the encounter record for clinical follow-up. This deployment does not alter the counselor's workflow — it augments it with objective data that the counselor can use or disregard based on clinical judgment.

6.4 Sports & Defense Operations

Real-time analysis of post-game interviews, press conferences, mission debriefs, and operational check-ins:

- **Post-Game Cognitive Assessment:** Athletes conduct post-game media sessions regardless of injury status. RTCA passively monitors these sessions for TBI markers (S7), cognitive decline (S4), and psychological distress (S1/S2) — providing a non-cooperative cognitive assessment from an interaction that already occurs.
- **Mission Debrief Analysis:** Military operators complete post-mission debriefs as standard procedure. RTCA analyzes these debriefs for TBI exposure markers, PTSD indicators, and operational readiness signals without requiring any additional assessment protocol.
- **AEGIS Live Certification:** Pilots, operators, and executives conduct routine check-in calls. RTCA converts these conversations into live AEGIS cognitive fitness assessments, enabling continuous certification monitoring through existing communication channels.

SECTION 07

Performance Specifications

7.1 Latency Requirements

Processing Stage	Target Latency	Acceptable Maximum
Audio Ingestion → Pre-Processing	<50ms	100ms
ASR (Speech-to-Text)	<500ms	800ms
Prosodic Feature Extraction	<100ms per 5s window	200ms
Lexical/Syntactic Features	<200ms per sentence	400ms
NOST Oscillatory Mapping	<150ms per window	300ms
NOSS Sector Scoring	<100ms per window	200ms
Alert Generation	<50ms	100ms

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End-to-End (Audio → Alert)	<2 seconds	3 seconds

7.2 Accuracy Specifications

Metric	At 60 Seconds	At 3 Minutes	At 10 Minutes	Batch Mode (Reference)
ASR Word Error Rate	8–12%	6–9%	5–7%	4–6%
Speaker Diarization Accuracy	85–90%	92–95%	95–98%	97–99%
NOSS Sector Concordance with Batch	75–82%	85–90%	92–96%	100% (baseline)
Prosodic Feature Reliability	88–93%	93–96%	96–98%	98–99%
Alert Sensitivity (RED/ORANGE)	85–90%	92–95%	95–98%	97–99%
Alert Specificity (RED/ORANGE)	80–85%	88–92%	93–96%	95–98%

Note: *Accuracy specifications are design targets for clinical validation. Actual performance will depend on audio quality, network conditions, speaker characteristics, and clinical context. All targets assume controlled acoustic environments (clinical setting, quiet office) with broadband or better audio quality. Degraded audio environments (speakerphone, mobile in noisy environment) will reduce accuracy across all metrics.*

7.3 Scalability Specifications

INFRASTRUCTURE SCALING TARGETS

Concurrent Streams (Initial)	**500 simultaneous calls**
Concurrent Streams (Year 2)	**5,000 simultaneous calls**
Concurrent Streams (Year 3)	**50,000 simultaneous calls**
Compute per Stream	**~0.5 vCPU + 0.25 GPU (inference)**
Bandwidth per Stream	**32 Kbps (feature-only); 256 Kbps (full audio)**
Storage per Call-Hour	**~120 MB (audio + features + results)**

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Regulatory and Privacy Framework

8.1 Consent Architecture

Real-time call analysis introduces heightened consent requirements compared to batch NOUS assessment, where the patient explicitly provides a sample. The RTCA Module implements a multi-layer consent framework:

- **Informed Consent (Clinical):** In clinical settings (telehealth, call center), patients receive explicit notification that cognitive-linguistic analysis will be performed during the call. Consent is obtained digitally prior to the call or verbally at the start of the call, with the consent event logged and timestamped.
- **Opt-Out Mechanism:** Patients can decline RTCA analysis at any time. The opt-out is immediate — all real-time processing ceases, the analysis window is cleared, and no partial results are retained.
- **Passive Monitoring Contexts:** In deployment scenarios where analysis is passive (e.g., employer wellness programs, sports media sessions), consent is obtained at enrollment rather than per-interaction. Participants are informed that their speech during designated interactions may be analyzed. All passive monitoring deployments require IRB or ethics committee review.
- **Two-Party vs. One-Party Consent:** RTCA deployment must comply with applicable wiretapping and recording consent laws, which vary by jurisdiction. In two-party consent states, all parties to the call must be informed. The system architecture supports per-jurisdiction consent configuration.

8.2 Data Handling

- **Audio Retention Policy:** Configurable per deployment. Options range from "feature-only" (audio is processed in real time and immediately discarded — only extracted feature vectors are retained) to "full retention" (encrypted audio stored for clinical review and quality assurance). The feature-only option is preferred for privacy-sensitive deployments.
- **PHI Segmentation:** Real-time transcripts and extracted features are stored separately from patient identifiers, linked only through tokenized references. De-identified feature vectors can be processed without PHI exposure.
- **Encryption:** AES-256 at rest for all stored data. TLS 1.3 for all in-transit data, including the audio stream, feature vectors, and alert messages. End-to-end encryption from the audio source to the processing pipeline when using the feature-only transmission mode.
- **Audit Trail:** Complete chain-of-custody logging for every data element: audio frame ingested, feature extracted, score computed, alert generated. Every processing event is timestamped and attributed.

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8.3 FDA Regulatory Pathway

The RTCA Module does not change the regulatory classification of the NOUS Platform. It remains Software as a Medical Device (SaMD) providing clinical decision support. The real-time delivery mechanism does, however, introduce additional regulatory considerations:

> **Regulatory Classification:** The RTCA Module produces the same clinical outputs (NOSS sector activations, trajectory alerts, differential classifications) as batch NOUS. The delivery mechanism (real-time vs. post-hoc) does not alter the intended use or the clinical decision support classification. FDA Class II (510(k)) pathway remains appropriate. However, the real-time alerting capability — particularly RED alerts for acute events — may attract additional scrutiny regarding time-critical performance validation. Clinical validation studies for RTCA will include specific latency and accuracy requirements for acute alert scenarios.

8.4 Ethical Considerations

- **Surveillance vs. Assessment:** Real-time analysis raises the ethical question of when clinical monitoring becomes surveillance. Beverly Index's position is clear: RTCA operates exclusively in contexts where clinical care is the purpose of the conversation and where explicit consent has been obtained. RTCA is not a covert monitoring tool. It is a clinical instrument deployed with transparency.
- **Alert Burden:** False positive alerts during live conversations carry a different clinical cost than false positives in batch mode. An incorrect RED alert during a telehealth visit may cause unnecessary alarm. Alert thresholds in RTCA mode are calibrated conservatively — higher specificity is preferred over higher sensitivity to minimize in-call disruption.
- **Clinician Override:** Every RTCA alert includes a one-click "Acknowledge/Dismiss" mechanism. The clinician retains full authority to override or ignore any system-generated alert. NOUS recommends; the clinician decides.
- **Equity in Audio Quality:** Patients calling from noisy environments, using older phones, or speaking with accents may experience degraded analysis accuracy. RTCA includes audio quality scoring that adjusts confidence levels based on signal quality, and the system transparently communicates reduced confidence to the clinician rather than producing potentially unreliable scores.

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SECTION 09

Development Roadmap

PHASE 1: STREAMING FOUNDATION Months 1–9

Focus: Core Streaming Pipeline & Prosodic Channel

- Develop streaming REMIEL adapter layer with windowed processing architecture
- Deploy real-time ASR engine with clinical vocabulary optimization
- Implement continuous prosodic feature extraction (F0, PVI, voice quality) at 5-second micro-window resolution
- Build speaker diarization module with voiceprint enrollment capability
- Develop edge-cloud hybrid architecture with feature-only transmission mode
- Internal validation: NOSS concordance testing between streaming and batch modes

PHASE 2: CLINICAL INTEGRATION Months 9–18

Focus: Telehealth Deployment & Clinical Validation

- Build WebRTC integration SDK for telehealth platform partners
- Develop live clinician dashboard with real-time NOSS visualization and alert overlay
- Implement real-time trajectory comparison engine (comparing live scores against patient longitudinal baselines)
- Deploy pilot with 3–5 clinical sites across mental health, cardiology, and oncology
- Commence clinical validation studies: streaming accuracy vs. batch accuracy, alert sensitivity/specificity
- Begin FDA pre-submission engagement for RTCA-specific performance claims

PHASE 3: SCALE DEPLOYMENT Months 18–30

Focus: Call Center Scale, Crisis Lines & Enterprise

- Deploy high-volume call center integration (SIP/VoIP tap architecture)
- Launch crisis line pilot with 988 network or regional crisis center partner
- Implement AEGIS live certification module for defense and aviation deployments
- Build sports media analysis pipeline for professional league partnerships
- Scale infrastructure to 5,000+ concurrent stream capacity
- Launch RTCA-specific per-minute and per-encounter pricing models
- Publish clinical validation results for FDA submission and peer-reviewed publication

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SECTION 10

Intellectual Property Considerations

The RTCA Module generates significant additional patentable innovation that extends Beverly Index's existing portfolio of 35 pending patent applications. Anticipated new patent filings include:

- Real-time streaming adaptation of oscillatory signature extraction from continuous audio input
- Windowed NOSS scoring methodology with confidence-weighted temporal accumulation
- Prosodic-autonomic real-time cardiac surveillance via speech rhythm analysis during live conversation
- Acute cerebrovascular event detection from real-time linguistic disruption pattern classification
- Speaker-aware multi-target cognitive assessment in multi-party clinical conversations
- Edge-cloud hybrid architecture for privacy-preserving real-time linguistic feature extraction
- Adaptive alert threshold calibration based on conversation duration and accumulated confidence

These filings would bring the total Beverly Index patent portfolio to an estimated 40–42 pending applications, further strengthening the company's defensive IP position in a market segment — real-time conversational cognitive analysis — where no other entity holds comparable patent protection.

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SECTION 11

Financial Overview

11.1 Revenue Model

Revenue Stream	Model	Price Point	Target Volume (Yr 3)
Telehealth Per-Encounter	Per-encounter fee	$3–$8/encounter	500,000+ encounters
Call Center Per-Minute	Per-minute processing	$0.08–$0.25/minute	10M+ minutes
Enterprise Platform	Annual license	$250K–$1.5M/year	10–20 systems
Crisis Line Licensing	Per-seat annual	$5K–$15K/seat	200–500 seats
Sports / Defense	Program license	$100K–$500K/year	15–30 programs

11.2 Projected Financial Impact

Metric	Year 1	Year 2	Year 3
Development Investment	$2.8M	$3.2M	$1.8M
Revenue (RTCA)	$0.5M	$4.8M	$14.0M
Gross Margin	−40%	38%	65%
Incremental Platform Value	$10M–$15M	$35M–$55M	$100M–$160M

Note: *These projections assume successful Phase 1–2 development and clinical validation milestones. Revenue estimates are conservative and assume modest market penetration in telehealth (0.2% of addressable encounters) and call center (0.1% of addressable minutes) by Year 3. Incremental platform value reflects RTCA's contribution to overall NOUS Platform enterprise valuation.*

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SECTION 12

Conclusion

The Real-Time Call Analysis Module represents the natural evolution of NOUS from an assessment tool to a surveillance instrument. The core science — Neural Oscillation Signature Theory, the REMIEL extraction pipeline, the NOSS classification system — is unchanged. What changes is the delivery modality: from samples collected and submitted, to conversations analyzed as they happen.

The technical architecture is sound. Streaming audio processing, windowed linguistic analysis, and real-time prosodic monitoring are well-understood engineering challenges with mature tooling available. The innovation is not in any individual component — it is in the application of these capabilities to the NOUS processing pipeline, enabling five-domain cognitive surveillance from live conversation.

The market timing is favorable. Telehealth is permanent. Call center AI is mature enough to support clinical-grade analysis. The ambient clinical intelligence market has established that clinicians will adopt conversational AI during encounters. Crisis lines are overwhelmed and actively seeking technological augmentation. Sports and defense organizations need real-time cognitive assessment that does not require cooperation or disruption.

The competitive position is unassailable. No platform in the ambient AI, call analytics, or clinical documentation space performs multi-domain cognitive-linguistic surveillance from live conversation. The 35+ patent portfolio protects the core methodology. The RTCA-specific patent filings will extend protection to the real-time deployment architecture. A competitor would need to independently develop equivalent science, build equivalent processing capability, and navigate equivalent regulatory pathways — against a portfolio that already exists and is expanding.

The RTCA Module converts every conversation into a clinical instrument. Every telehealth visit, every phone call, every check-in becomes a five-domain cognitive surveillance session — silent, objective, and continuous. The infrastructure already exists. The conversations are already happening. NOUS simply starts listening.

Beverly Index LLC

Comprehensive Clinical Cognitive Surveillance

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